

MAHARAJA P

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EDUCATION

B.E., Electronics & Communication Engineering — SSN College of Engineering, Chennai

2024 – 2028 (Pre-Final Year) | CGPA: 8.104

OBJECTIVE

Pre-final year Electronics & Communication Engineering student pursuing the Technology Developer (Summer Intern) role at Barclays. Backend-focused engineer with hands-on experience designing REST APIs, multi-threaded services, and secure, scalable systems using Java and Spring Boot, combined with a track record of end-to-end ownership, clear cross-functional communication, and disciplined secure coding practices.

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Multi-threaded Programming, System Design Fundamentals, SDLC

Languages: Java (Core & OOP, Multithreading), SQL, JavaScript/GUI (Foundational)

Backend & APIs: Spring Boot (Embedded Tomcat), REST API & Microservice Development, JWT Authentication & Role-Based Access Control, BCrypt Encryption, Secure Coding Practices

Databases: MySQL — Joins, ACID, Transactions, Schema Design

Tools & Deployment: Git/GitHub, Maven, Postman (API Testing), JUnit, IntelliJ IDEA, Docker, AWS, Vercel, Railway, Render

EXPERIENCE

Zenith Hackathon (January 2026) — Identified a real, recurring problem (employees struggling to locate stock in front of customers), built Store Lens for the hackathon, engaging directly with the product owner (store owner) and staff to gather requirements, and optimized the system based on that feedback.

Sustainability Hackathon (May 2026) — Participated in design sessions to architect the underground water leak-detection system (below), then presented it to judges and US-based sponsors, communicating system design and demo results; collaborated on funding and scalability feedback.

PROJECTS

Smart Underground Water Distribution & Leak Detection System

github.com/pmaharajacs7-hue/Water_Leakage_Detection_System

Team of 2 | Backend Engineer • Java (Spring Boot), MongoDB, Python, REST APIs, IoT (ESP32)

- Built a low-cost, easy-to-operate leak-detection system for underground pipelines, aimed at villages and municipalities priced out of expensive, complex enterprise tools.
- Designed the Spring Boot backend and REST APIs to ingest live ESP32 sensor data in real time, combined with EPANET hydraulic modeling and a Python ML anomaly-detection layer to flag leaks by location.
- Applied statistical feature engineering (rolling averages, standard deviation) to cut false alarms, resolving cross-stack integration issues across Java, Python, and IoT through systematic debugging.
- Resolved a shared-state race condition across concurrent per-pipe sensor streams by refactoring to a thread-safe ConcurrentHashMap, ensuring accurate rolling statistics under multi-threaded load.

Store Lens — Smart Store Management System

github.com/pmaharajacs7-hue/Store-Lens | store-lens-r6l8.onrender.com

Team of 2 | Backend Engineer • Java, Spring Boot, MySQL, JWT, REST APIs, OpenAI Whisper API

- Built a Java/Spring Boot inventory system so any employee can instantly locate stock — solving the delays new hires faced searching for items in front of customers.
- Hand-built the backend and REST APIs end-to-end; added the OpenAI Whisper API for voice search and JWT-based role access control (AI-assisted) to block cross-shop data access.
- Automated bulk inventory updates via ACID-compliant CSV upload and added purchase-trend visibility for owners, cutting new-employee onboarding time.
- Deployed and maintained the application on Render (production hosting); diagnosed and resolved deployment failures — including environment variable misconfigurations and database connection issues — by analyzing application logs to restore uptime.

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

Lead Volunteer, National Service Scheme (NSS), SSN Chapter: Led environmental cleanup drives, rural outreach, and Blood Donation/Voting Awareness camps.

Student Volunteer, Talent Quest for India (TQI) – Chengalpattu Cluster II: Delivered an 8-week educational curriculum for grades 9–12, building public speaking and mentoring skills.

Software Coordinator, Department of ECE, SSN College of Engineering: Bridged hardware curricula and software practice through coding resources and peer mentorship.

ADDITIONAL INFORMATION

Languages: Tamil (Native), English (Professional Proficiency)

Soft Skills: Cross-Functional Collaboration, Secure Coding Mindset, Results-Oriented Self-Starter, Code Reviews & Knowledge Sharing